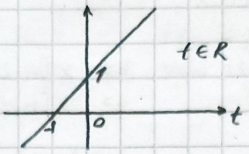
Sinyaller (İşaretler) => 1- Sürekli zaman sinyalleri
2- Ayrık zaman sinyalleri

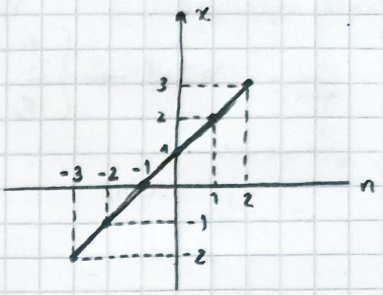
$y = f(x) = x^2 + 1$
$x(t) = t + 1$

1- $x(t) = t + 1$

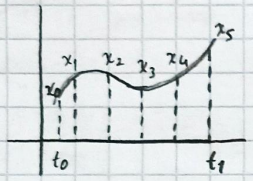


Sürekli
(fonksiyon)

2- $x[n] = n+1$, $n \in Z$

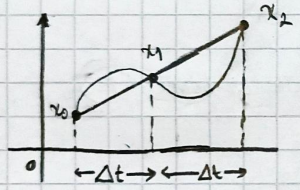


Ayrık
(Dizi)



örnekleme (sampling)
$\{x_0, x_1, x_2, x_3, x_4, x_5\}$

ör: $x[n] = \{-1, 0, 3, 5\}$



$\longleftrightarrow$

$T_{sampling} \propto f_s$ => Δt daha küçükse daha fazla bölme alınır, daha kaliteli ses olur ama daha fazla yer alır.

$f_s = \frac{1}{T_s}$

Δt büyükse az bölme alınır, kalitesiz ses olur ama daha az yer alır.

f_s = 8 kHz → saniyede 8 bin örnek
f_s = 48 kHz → // // 48 bin örnek
↳ daha kaliteli

Δt sayısı yüksek; kaliteli ses yüksek depo

Δt sayısı az; kalitesiz ses az depo

f_s = 16kHz → 8 kHz'e kadar duyulan sesleri kaydeder (yarısı)

Not: İnsanlar 0-20 kHz arası sesleri duyar.

Nyquist Kriteri

$f_s \geq 2f \Rightarrow f \leq \frac{f_s}{2}$

örnekleme frekansı, örneklenen frekanstan en az 2 kat büyük olmalı.

Sinyallerin Dönüşümü

zamanda kaydırma $x(t) \longrightarrow x(t \mp t_0)$, $x[n] \longrightarrow x[n \mp n_0]$

zamanda ölçekleme $x(t) \longrightarrow x(a.t)$, $x[n] \longrightarrow x[a.n]$

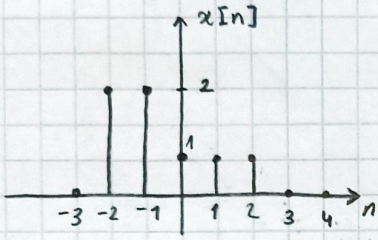
zamanda ters çevirme $x(t) \longrightarrow x(-t)$:

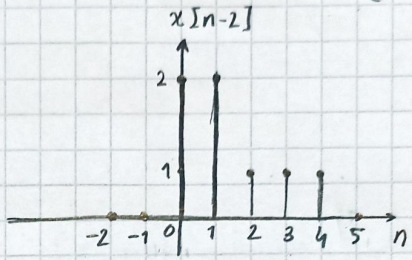
Genlik ölçekleme $x(t) \longrightarrow a.x(t)$:

örnek: $x[n] \rightarrow \begin{cases} 1, & 0 \le n \le 2 \\ 2, & -2 \le n \le -1 \\ 0, & \text{diğer} \end{cases}$

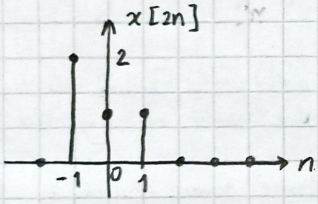
$\Rightarrow x[n-2] = ?$

$\Rightarrow x[n-2] = \begin{cases} 1, & 0 \le n-2 \le 2 \\ 2, & -2 \le n-2 \le -1 \\ 0, & \text{diğer} \end{cases} \Rightarrow \begin{cases} 1, & 2 \le n \le 4 \\ 2, & 0 \le n \le 1 \\ 0, & \text{diğer} \end{cases}$



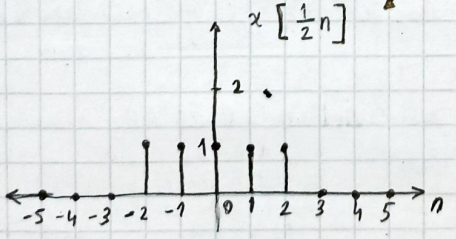


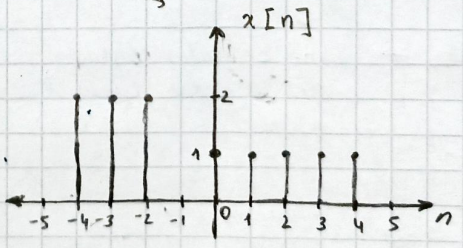
- $x[2n] = \begin{cases} 1, & 0 \le 2n \le 2 \\ 2, & -2 \le 2n \le -1 \\ 0, & \text{diğer} \end{cases} = \begin{cases} 1, & 0 \le n \le 1 \\ 2, & -1 \le n \le -\frac{1}{2} \\ 0, & \text{diğer} \end{cases}$



($-\frac{1}{2}$ tam sayı değil, ondan küçük en büyük -1)

- $x[\frac{1}{2}n] = \begin{cases} 1, & 0 \le \frac{1}{2}n \le 2 \\ 2, & -2 \le \frac{1}{2}n \le -1 \\ 0, & \text{diğer} \end{cases} = \begin{cases} 1, & 0 \le n \le 4 \\ 2, & -4 \le n \le -2 \\ 0, & \text{diğer} \end{cases}$





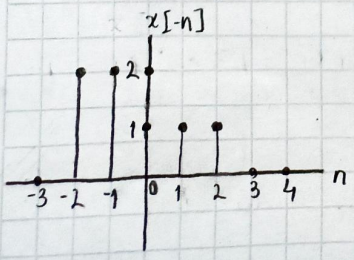
$x[-n] = \begin{cases} 1, & 0 \le -n \le 2 \\ 2, & -2 \le -n \le -1 \\ 0, & \text{diğer} \end{cases} = \begin{cases} 1, & 0 \ge n \ge -2 \\ 2, & 2 \ge n \ge 1 \\ 0, & \text{diğer} \end{cases}$

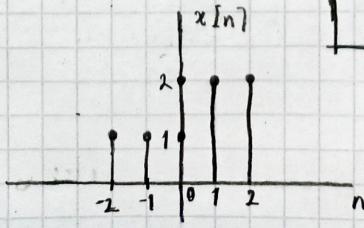
$x[n]$ tablosundan kontrol edersek

$x[\frac{1}{2}n] = x[0] = 1$

$n = -3$, $n = 1$, $n = 3$ değerlerinin yanlış yapıldığını buluruz

çünkü $n \in \mathbb{N}$ olmalı





Buraya bak

* $x[2n-1]$ ← $x[2n-1]$ $n=2n$

$x[n]$ → $n=n-1$ → $x[n-1]$

önce zamanda kaydırma
sonra ölçekleme

$x[2n-1]$
$n=2n$
$x[2n]$
↓
tersi nedir?
1/2 ile genişlet
$(n-\frac{1}{2})=n$

$2n-1 = 9/6$

$\frac{1}{2n-1} =$

$x[n-a]$	$x[-n+a]$
$a>0$ → sağa kayar	$a>0$ → sola kayar
$a<0$ → sola kayar	$a<0$ → sağa kayar

$x[-n+2]$

$x[n]$
↓
$n=-n$ yansıma
$x[-n]$
↓
$n=n-2$
$x[-n+2]$

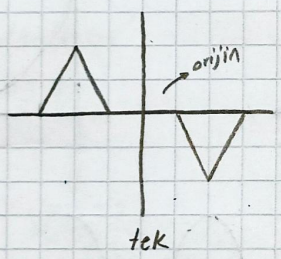
Çift Sinyal (even)

$x_{even}[n] = x_e[n]$

Tek Sinyal (odd)

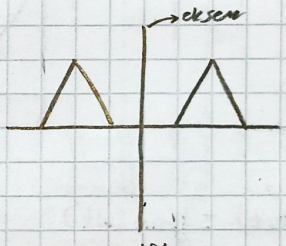
$x_{odd}[n] = x_o[n]$

$x[n] = x_e[n] + x_o[n]$



tek

$x[n] = -x[-n]$



çift

$x[n] = x[-n]$

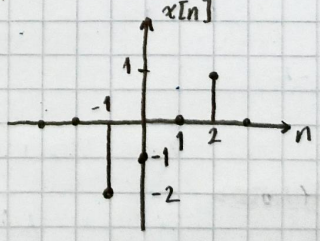
$x_o[n] = \frac{x[n]}{2} - \frac{x[-n]}{2} = x[n]$

$x_e[n] = \frac{1}{2}\left(x[n] + x[-n]\right)$

$x_o[n] = \frac{1}{2}\left[x[n] - x[-n]\right)$

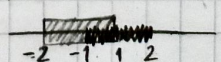
n	$x[n]$	$x[-n]$	$x_e[n]$	$x_o[n]$
-2	$x[-2]=0$	$x[2]=1$	$x_e[2]=1/2$	$x_o[2]=-1/2$
-1	$x[-1]=-2$	$x[1]=0$	$x_e[-1]=-1$	$x_o[-1]=-1$
0	-1	-1	-1	0
1	0	-2	-1	1
2	1	0	1/2	1/2

örn: $x[n] = n-1$, $-1 \le n \le 2$



$x_e[n] = \frac{1}{2}\left(x[n] + x[-n]\right) = \frac{1}{2}\left(n-1 + (-n-1)\right)$

$x[-n] \Rightarrow 1 \ge n \ge -2 \quad = -1$

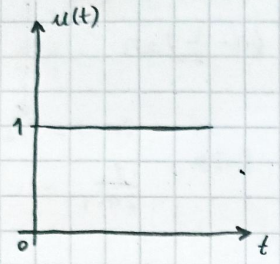


$x_o[n] = n$

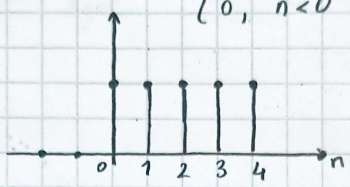
Temel Fonksiyonlar ~ Sinyaller

1- Birim Basamak (Unit Step) : $u(t)$, $u[n]$

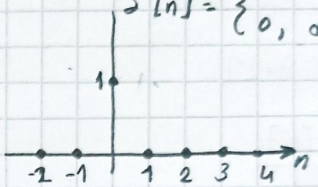
2- Birim Dürtü (Impulse) $\delta(t)$ → sürekli zamanda , $\delta[n]$ → ayrık zamanda



$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

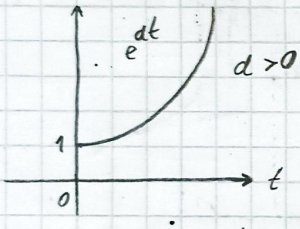


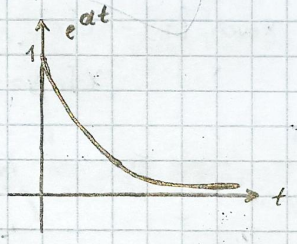
$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & \text{diğer} \end{cases}$$

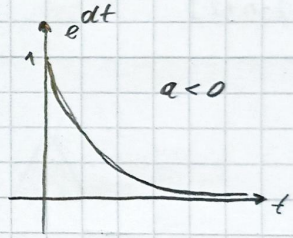


1- Üstel , $e^{(\alpha + j\omega)t}$

2- Sinüzodialler , $\cos(\Omega n + \varphi)$, $\sin(\Omega n + \varphi)$ → ayrık zamanda



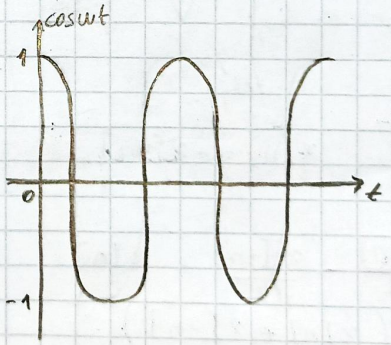




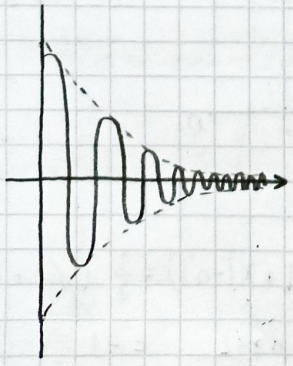
$e^{\pm j\omega} \Rightarrow \cos(\omega t) \pm j\sin(\omega t)$

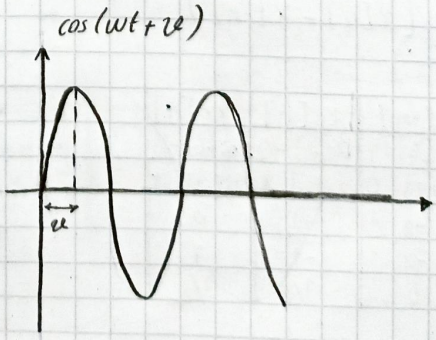
$\cos\omega t \Rightarrow \dfrac{e^{j\omega t} + e^{-j\omega t}}{2}$

$\sin\omega t \Rightarrow \dfrac{e^{j\omega t} - e^{-j\omega t}}{2j}$

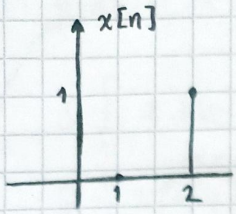


$\{e^{at} \cdot e^{j\omega t}\}$

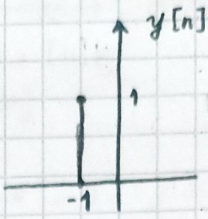




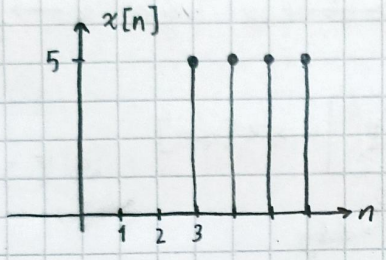
- $x[n] = \delta[n-2]$



- $y[n] = \delta[n+1]$



- $x[n] = 5u[n-3]$



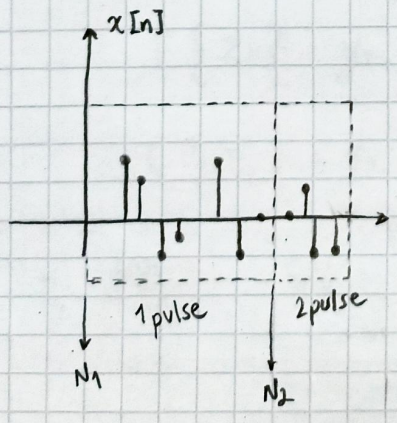
- $y[n] = x[2n-1]$

Enerji ve Güç Hesaplamaları

$$E = \sum_n |x[n]|^2$$

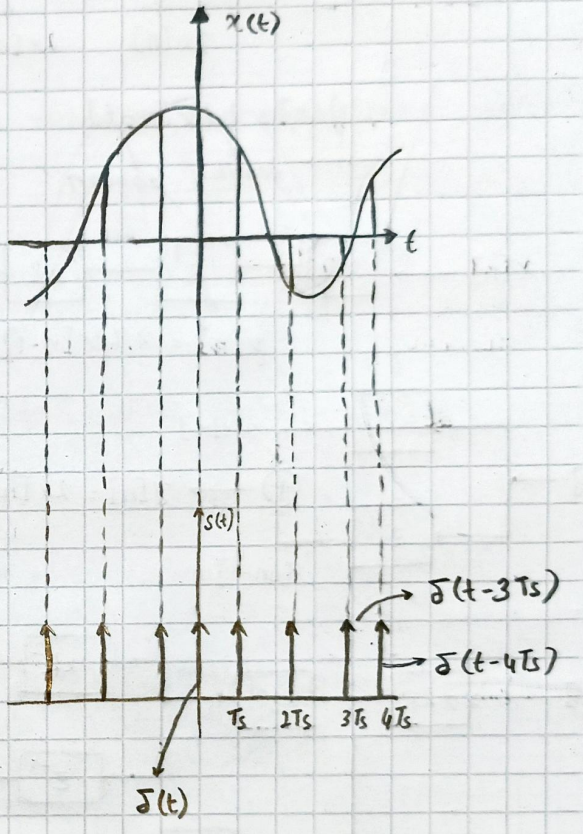
$$p = \text{mean}(E) = \frac{E}{N_2 - N_1 + 1}$$

Birim zamanda ortalama harcanan enerji



NYQUIST KRİTERİ

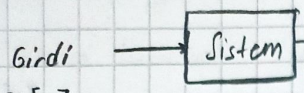
$2f \leq f_s$



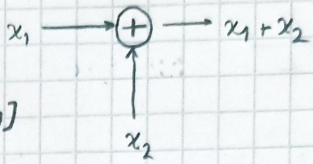
$$s(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT_s)$$

$x[n] = \{ \ldots \}$

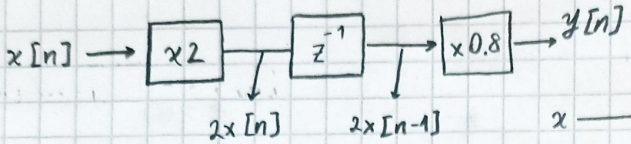
Sistem

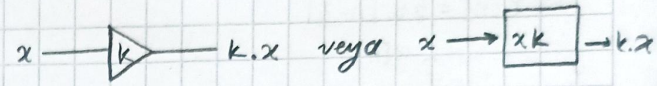


$y[n] = 2x[n] - x[n-1]$



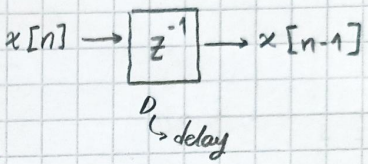
örnek :

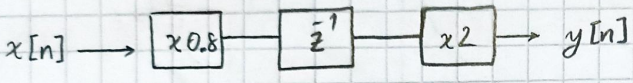




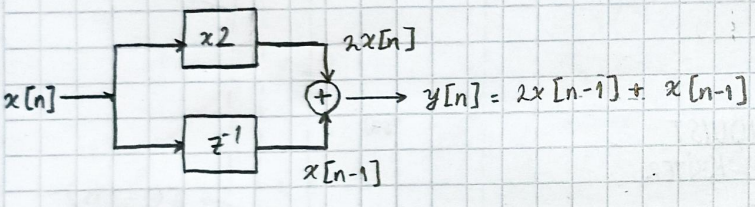
$y[n] = 1.6x[n-1]$

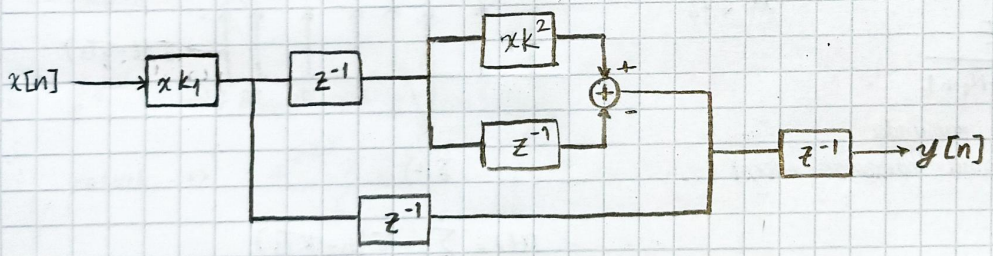
genlik — zaman

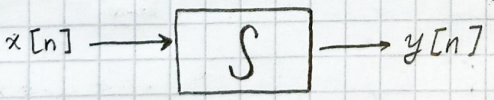




$y[n] = 1.6x[n-1]$





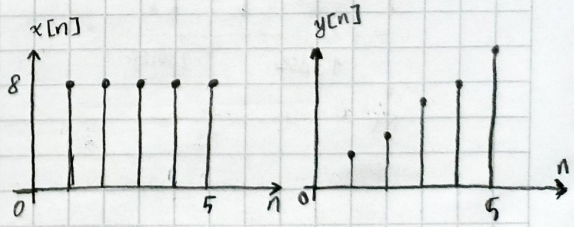


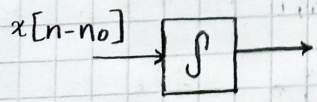
$S\{x[n]\} = y[n]$

Doğrusal : Linear

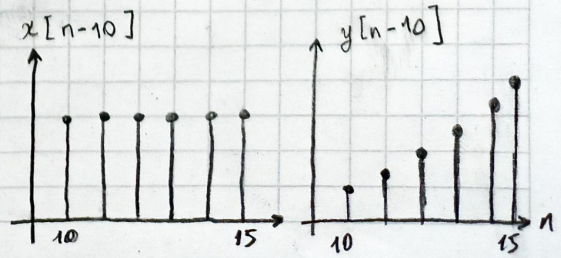
Zamanla Değişmeyen – Time Invariance

Doğrusal Zamanla Değişmeyen Sistemler – LTI





$S\{x[n-1]\} = y'[n]$



$S: y[n] = 2x[n-1]$ bu sistem, zamanla değişmeyen bir sistem midir?

$S\{x[n]\} = 2x[n-1] = y[n]$

$S\{x[n-1]\} = 2x[n-1-1] = 2x[n-2] = y'[n]$

$y[n-1] = 2x[n-1-1] = 2x[n-2]$

$y'[n] = y[n-1]$

Eşit olduğu için zamanla değişmeyen (Time Invariance) bir sistemdir?

- $y[n] = e^{-2n}x[n]$

} zamanla değişmeyen sistemler olup olmadıklarını göster?

Doğrusallık - Lineerlik

Bir sistemin lineer olabilmesi için iki koşul sağlanmalıdır;

Homojenlik + Toplamsallık (Süperpozisyon)

$x[n] \rightarrow y[n]$
$a.x[n] \rightarrow a.y[n]$

} Homojenlik Koşulu,

a / $x_1[n] \rightarrow y_1[n]$
b / $x_2[n] \rightarrow y_2[n]$

$a.x_1[n] + b.x_2[n] \rightarrow a.y_1[n] + b.y_2[n]$

} Toplamsallık Süperpozisyon (superposition) koşulu

$S\{x_1\} = y_1$, $S\{a.x_1\} = a.y_1$

$S\{x_2\} = y_2$, $S\{b.x_2\} = b.y_2$

, $S\{a.x_1 + b.x_2\} = a.y_1 + b.y_2$

- $y[n] = x[n] + 2x[n-2] = S\{x[n]\}$

Girdi

$x_1[n] \Rightarrow S\{x_1[n]\} = y_1[n] = x_1[n] + 2x_1[n-2]$

$a.x_1[n] \Rightarrow S\{a.x_1[n]\} = a.x_1[n] + 2a.x_1[n-2] = a.y_1[n]$
$b.x_2[n] \Rightarrow S\{b.x_2[n]\} = b.x_2[n] + 2b.x_2[n-2] = b.y_2[n]$

$a.x_1[n] + b.x_2[n] \Rightarrow S\{a.x_1[n] + b.x_2[n]\} = a.x_1[n] + b.x_2[n] + 2[a.x_1[n-2] + b.x_2[n-2]]$
$= a.y_1[n] + b.y_2[n]$ olmalıdır.

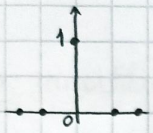
$a.y_1[n] + b.y_2[n] = a.(x_1[n] + 2x_1[n-2]) + b.(x_2[n] + 2x_2[n-2])$

örnek : $y[n] = n.x[n]$, LTI ?

- Birim Dürtü Tepkisi (Impulse Response):

$x[n] \longrightarrow \boxed{h[n]} \longrightarrow y[n]$ $\quad \Rightarrow \quad y[n] = x[n] * h[n]$

$\delta[n] \longrightarrow \boxed{?} \longrightarrow h[n]$



birim dürtü, 0 anında 1 geri 0.

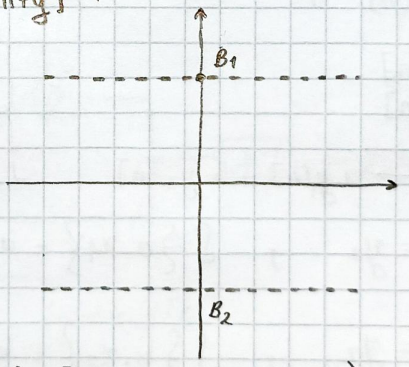
- Nedensellik

$h[n] = 0$ for $n < 0$



- Kararlılık (Stability) - [BIBO Stability] :

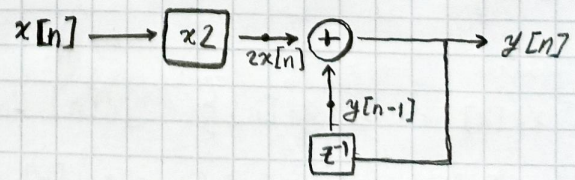
$\sum_n |h[n]| < \infty$



LCDDE (Linear Constant - Coefficient Difference Equations)

$y[n] = y[n-1] + 2x[n]$

Blok Şema



$x[n] = \delta[n] \Rightarrow y[n] = h[n]$

n	$\delta[n]$	$y[n] = h[n] = y[n-1] + 2\delta[n]$	
0	$\delta[0] = 1$	$y[0] = y[-1] + 2\overbrace{\delta[0]}^{2.1} = 2$	$h[n] = 2$
1	$\delta[1] = 0$	$y[1] = \overbrace{y[0]}^{2} + 2\overbrace{\delta[1]}^{0} = 2$	
2	$\delta[2] = 0$	$y[2] = \overbrace{y[1]}^{2} + 2\overbrace{\delta[2]}^{0} = 2$	

örnek : $y[n$